

Routing Protocol Design

EN2150 Communication Network Engineering
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Intended Learning Outcomes

1. LO1: Identify and describe the core operational principles of key internet routing protocols (RIP, OSPF, IS-IS, and BGP), and analyze their limitations in terms of scalability, convergence, fault tolerance, and security.
2. LO2: Design and develop an improved or novel routing protocol that addresses identified limitations, applying appropriate network design principles and using simulation tool.
3. LO3: Evaluate and compare the performance of the proposed routing protocol against existing protocols through simulation or prototyping, and communicate findings effectively through technical reports and presentations

Background

Current routing protocols like RIP (which uses distance vectors), OSPF and IS-IS (which use link-state information), and BGP (used between different networks) have made internet communication possible on a large scale. But they also have some limitations such as being slow to update, not handling very large networks well, having security issues, and not reacting quickly to changes in the network.

Objective

Design a new routing protocol that addresses at least two significant limitations of the existing protocols (RIP, OSPF, IS-IS, and BGP). Your design should consider scalability, efficiency, fault tolerance, convergence behavior, and security¹. Your new design should clearly highlight what changes you are making, why they are necessary, and how they improve scalability, efficiency, fault tolerance, convergence behavior, or security

¹You may choose to use an existing routing protocol (such as RIP, OSPF, IS-IS, or BGP) as a baseline and propose improvements or extensions to address its limitations.

Assignment Components and Team Roles

Each team member should take on one of the following roles. While primary responsibilities are assigned, collaboration and task-sharing across roles (specially protocol design) are strongly encouraged to ensure that all members develop a well-rounded understanding of the entire protocol design process. In the final report, include a section that clearly outlines each team member's contributions, including both their assigned role and any additional tasks they supported with percentages.

1. Literature Review

Task: Study the current limitations of RIP, OSPF, IS-IS, and BGP.

Deliverable: Summary document highlighting key weaknesses of each protocol with examples and references.

2. Protocol Design

Task: Design the core of the new protocol. Decide on:

- Routing metric(s)
- Control message format
- State information stored at routers
- Algorithm for route computation

Deliverable: Technical design document (including flow diagrams and pseudocode).

3. Simulation

Task: Implement a basic version of the protocol using simulation tools (e.g., OMNeT++, Python with NetworkX, NS3, Mininet).

Deliverable: Code + Simulation report showing performance compared to at least one existing protocol.

4. Security & Scalability Analyst

Task: Evaluate the new protocol's scalability and resistance to common attacks (e.g., route hijacking, spoofing, loops).

Deliverable: Analytical report including test scenarios, failure points, and suggested improvements.

Final Deliverables

1. Report

- Introduction
- Summary of existing protocols & their limitations
- Proposed protocol design
- Performance analysis

- Security & scalability analysis
- Conclusion

2. Simulation Code

- Annotated source code
- README file with instructions to run the code

In your simulation, you are required clearly demonstrate at least one specific weakness of an existing routing protocol. Then, simulate the same scenario using your proposed protocol to show how it improves on that issue. Provide quantitative or visual evidence (e.g., graphs, tables, or logs) to support your comparison and explain why your design performs better.

Evaluation Criteria

Total mark allocation 25% of overall course marks

Criteria	Percentage
Depth of analysis of existing protocols	20%
Technical correctness of the proposed protocol	25%
Originality & practicality of the proposed protocol	25%
Clarity of documentation	10%
Quality of simulation	20%

Submission

- Upload a report as a PDF file named as "EN2150_your_groupno_A04.pdf". Include the index numbers and the names of the group members within the report. Make sure to include a link in the report to download your source code or simulation files. Only one submission is required from a group.
- The interpretation of results and the discussion are important in the report.
- Plagiarism will be checked and in cases of plagiarism, an extra penalty of 50% will be applied. In case of copying from each other, both parties involved will receive a grade of zero for the assignment. Academic integrity is of utmost importance, and any form of plagiarism² or cheating will not be tolerated.
- An extra penalty of 15% is applied for late submission.

²<https://en.wikipedia.org/wiki/Plagiarism>

References

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